

Class introduction

Ryan Giordano

Key concepts

This class will focus almost entirely on *supervised learning*.

Here are the key concepts:

- ▶ What does it mean to do well on a supervised learning problem?
 - ▶ Random data, loss, and risk
 - ▶ **Empirical risk minimization, generalization**
- ▶ What would we do if we had an infinite amount of data?
 - ▶ Generating computational tools to fit arbitrary functions
 - ▶ **Expressive function classes**
- ▶ What goes wrong when we have only a limited amount of data?
 - ▶ Avoid overfitting by limiting the expressivity of function classes
 - ▶ **Regularization**
- ▶ How do we find a sweet spot between expressiveness and overfitting?
 - ▶ Theory: **generalization bounds**
 - ▶ Practice: **out-of-sample error**

Day one walkthrough

- ▶ Fitting a given function if all you can do is query it
 - ▶ Function classes
 - ▶ What if you need to make a prediction at a point you've never seen before?
 - ▶ An approximation scheme comes with a class of functions that it can represent.
 - ▶ Loss
 - ▶ You've chosen some approximation. How well does it do?
 - ▶ When you observe the function exactly, is there any reason not to choose an arbitrarily expressive function class?
 - ▶ Plot best function in class versus truth as function class increases
 - ▶ Randomly chosen query locations
 - ▶ Where should you query?
 - ▶ You need to know that future queries will need to look like the present queries somehow
 - ▶ Risk (versus pointwise loss)
 - ▶ Can you guaranteed low loss on a point you've never seen before? (Spiky functions)
 - ▶ Assumptions about confounders

Day two walkthrough

- ▶ Random responses
 - ▶ Conditional distributions of responses
 - ▶ What happens if you query the same point twice and get different answers? Conditional distribution
 - ▶ Irreducible loss
 - ▶ What's the best you can do if there's noise in the response?
 - ▶ Optimal prediction function, and gap with function class
 - ▶ Overfitting
 - ▶ Suppose you have chosen a nice, expressive class, but have random y .
 - ▶ Can you tell whether you've overfit just by looking at the data? (No)
 - ▶ Plot best function in class versus truth vs data
 - ▶ Checking for overfitting for a single x
 - ▶ For a given point, ideally check the same point multiple times
 - ▶ For a given point, check nearby points and assume smoothness
 - ▶ Checking for overfitting when risk is the goal
 - ▶ If all we want is risk, get a new point to check.
 - ▶ And if your data is IID and you can't get a new point, just leave a point out
 - ▶ If the function class is smooth, and you can "borrow" from neighbors, you might hope that the estimate goes to the best function in the class

QSAR Fish Toxicity Dataset

These examples will be based on the UCI QSAR Fish Toxicity Dataset.

The dataset consists of a set of 908 chemicals (one row each), with data for:

- ▶ LC50: the concentration that causes death in 50% of test fish over a test duration of 96 hours.
- ▶ MLOGP: (molecular properties)
- ▶ CIC0: (information indices)
- ▶ GATS1i: (2D autocorrelations)
- ▶ NdsC: (atom-type counts)
- ▶ NdsCH: (atom-type counts)
- ▶ SM1_Dz: (2D matrix-based descriptors)



Figure 1: Sad fish

“Quantitative structure–activity relationship” (QSAR) models predict some chemical property (toxicity to the fathead minnow) from structural features of the molecules.

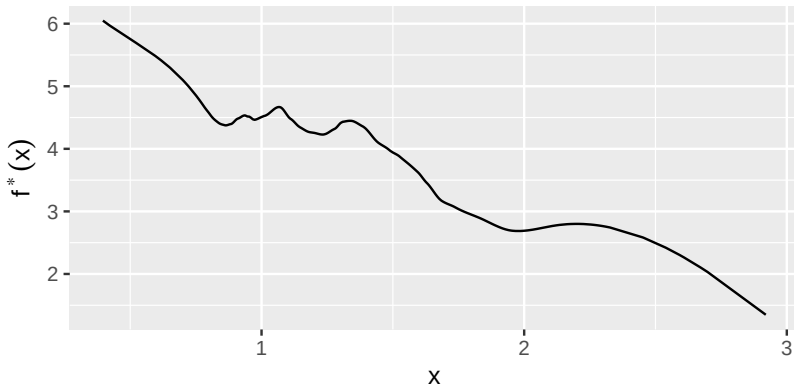
Task: Set $x = \text{GATS1i}$, $y = \text{LC50}$, and use ML to predict fish mortality from 2D autocorrelations.

i Question

Describe some concrete settings when you would want to use an ML model that could perform this task.

Note: I will be simulating data using a “ground truth” that is loosely based on patterns from the actual data.

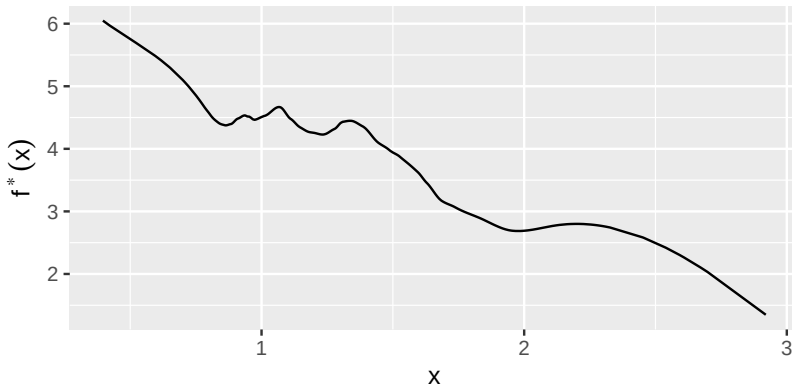
Fitting a function



Let's imagine that y (fish toxicity) is determined exactly by x (2D autocorrelations).

That means $y = f^*(x)$ for some f^* .

Fitting a function

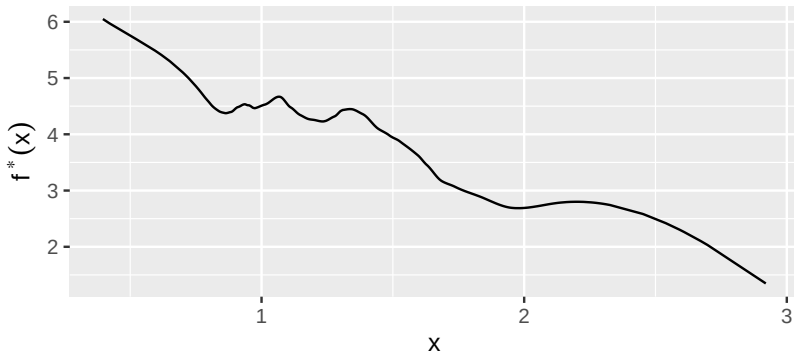


Let's imagine that y (fish toxicity) is determined exactly by x (2D autocorrelations).

That means $y = f^*(x)$ for some f^* .

We don't know f^* , but we can *query* it by selecting x (a molecule) and measuring y (seeing what concentration kills minnows)

Fitting a function



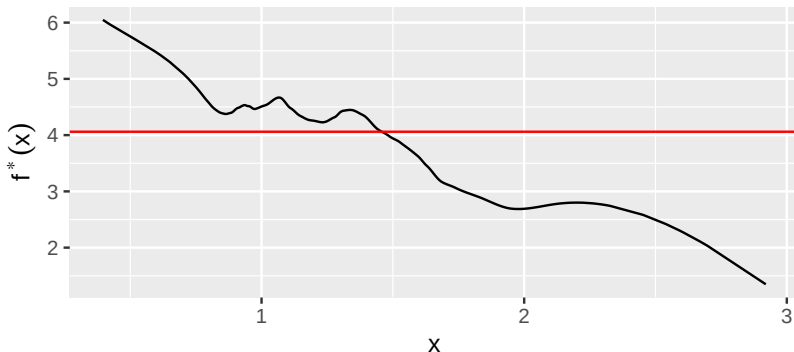
Task: Query (x_n, y_n) pairs and make a *program* that returns $\hat{f}(x) \approx f^*(x)$.

Question

Suppose you can query a very large number of datapoints.

- ▶ Can you ever perfectly approximate $f^*(x_{\text{new}})$ at an unseen x_{new} ?
- ▶ Propose the simplest procedure you can think of (even if not very good).
- ▶ Propose a more complex procedure that will do a better job.

Constant functions

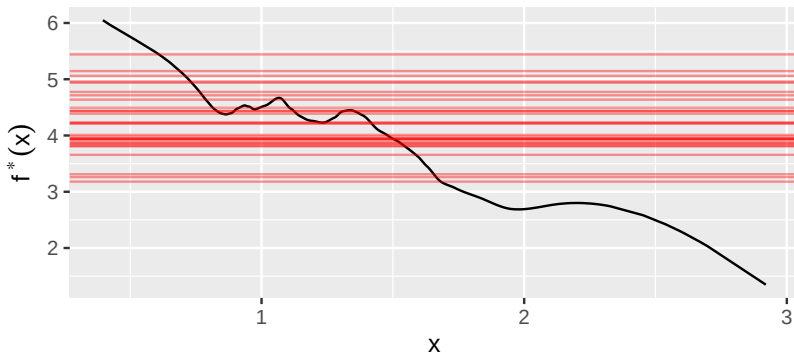


The simplest approximation is arguably the constant $\hat{f}(x) = \frac{1}{N} \sum_{n=1}^N f^*(x_n)$.

Question

- ▶ Think of an application for which a good constant is enough.
- ▶ Think of an application for which a good constant is not enough.
- ▶ Will the constant you estimate depend on which x_n you select? How?
- ▶ Can you think of other ways to take the data and choose the constant?

Constant functions



Using a constant function selects $\hat{f}$ from the *function class*:

$$\mathcal{F}_{constant} := \{f : f(x) = c \text{ for some real-valued } c\}$$

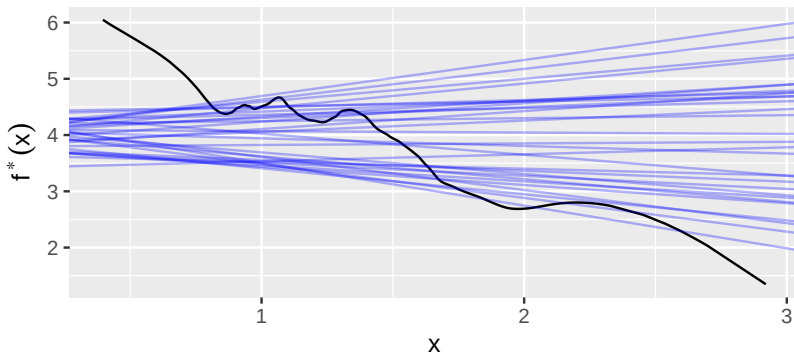
The function class is the set of *all functions that you could represent* with your procedure.

An “estimation procedure” takes in data $\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$ and returns a function $\hat{f} \in \mathcal{F}_{constant}$.

This is equivalent to choosing a $\hat{c}$ and taking $\hat{f}(x) = \hat{c}$.

This class is about designing and studying estimation procedures.

Linear functions



We can also define the class of linear functions:

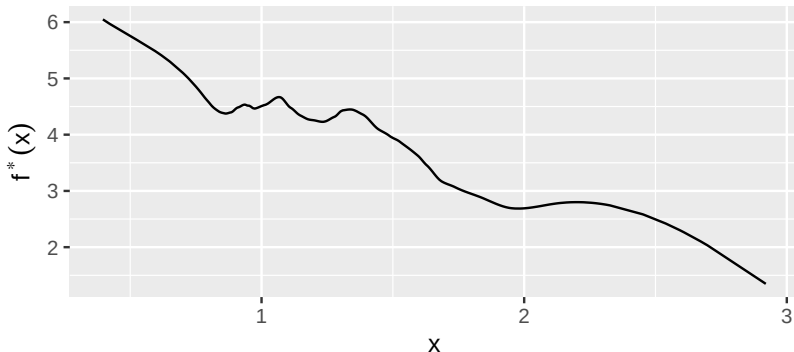
$$\mathcal{F}_{constant} := \{f : f(x) = c \text{ for some real-valued } c\}$$

$$\mathcal{F}_{linear} := \{f : f(x) = c + bx \text{ for some real-valued } c, b\}$$

Every constant function is also a linear function, but some linear functions are not constant functions.

We say that $\mathcal{F}_{linear}$ is a *more expressive function class* than $\mathcal{F}_{constant}$.

Function classes

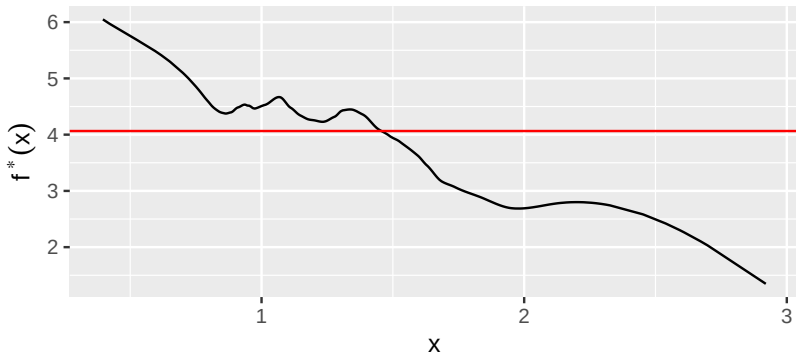


A key theme of this class is developing expressive function classes!

i Question

Identify the function classes for some of the examples given earlier. Which are more expressive? Which are the least expressive?

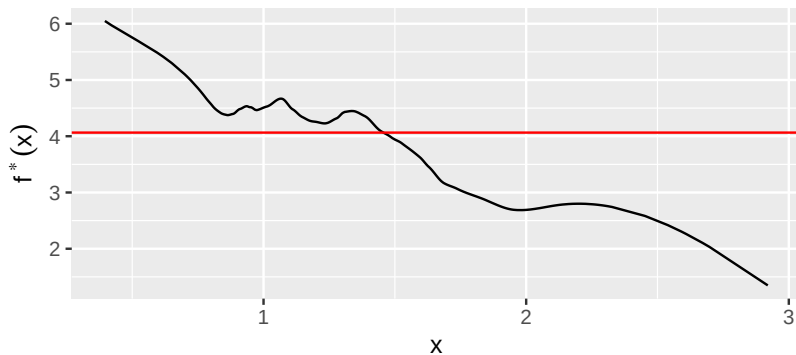
Loss



We want to choose $\hat{f} \in \mathcal{F}$ that is a “good approximation,” so we have to say what “good” means.

Suppose we have some x with associated y . Let the “loss function” $\mathcal{L}(f(x), y)$ measure how “bad it is” to predict $f(x)$ when the truth was y .

Loss



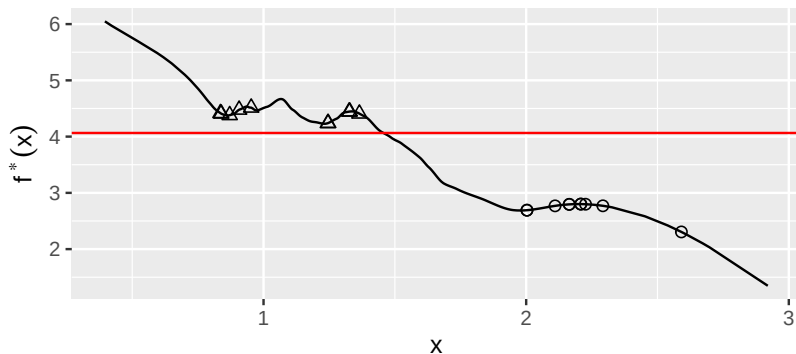
Common examples:

- ▶ $\mathcal{L}(f(x), y) = (f(x) - y)^2$
- ▶ $\mathcal{L}(f(x), y) = |f(x) - y|$
- ▶ $\mathcal{L}(f(x), y) = I(f(x) \neq y)$ (Here, $I(\cdot)$ is the “indicator function.” It evaluates to 1 if its argument is true, and 0 otherwise.)

Question

What are some properties any loss function should satisfy?

Risk

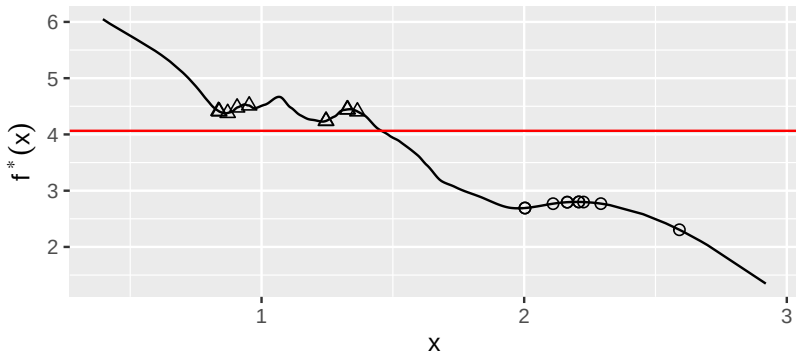


The loss is defined at a single datapoint. It makes sense to *average* the loss over multiple datapoints:

$$\frac{1}{N} \sum_{n=1}^N \mathcal{L}(f(x_n), y_n)$$

Note that, for a given f , the average loss will be different depending where the points are! **Where should you evaluate your loss?**

Risk



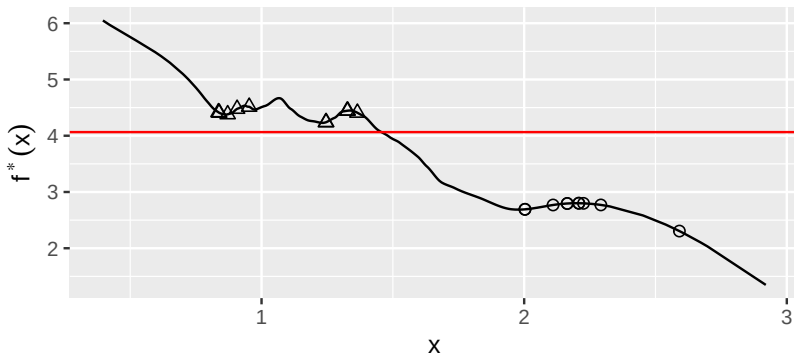
If we believe that future x will be drawn randomly, independently, and identically distributed (IID) from a future distribution $p(x)$, then it makes sense to measure the *average loss*.

This average loss is called “risk”.

$$\mathcal{R}(f) := \mathbb{E}_{p(x,y)} [\mathcal{L}(f(x), y)] \quad \text{”Risk”}$$

Note that risk depends only on the function — the distribution or data is considered fixed.

Risk



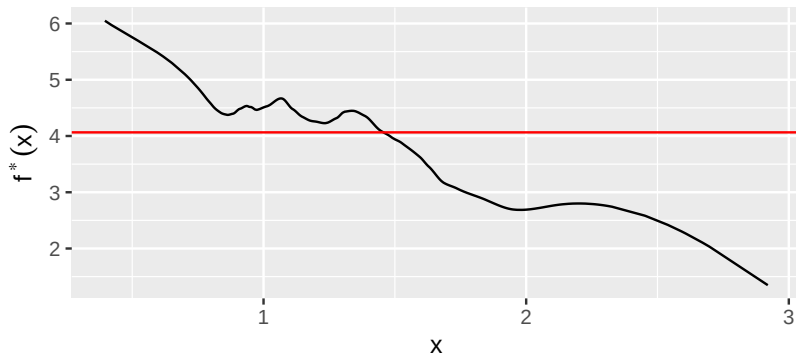
If we want to learn a function that makes $\mathcal{R}(f)$ small, we should draw training data $x_n \stackrel{\text{iid}}{\sim} p(x)$ and form the “empirical risk:”

$$\mathcal{R}(f) := \mathbb{E}_{p(x,y)} [\mathcal{L}(f(x), y)] \quad \text{”Risk”}$$

$$\hat{\mathcal{R}}(f) := \frac{1}{N} \sum_{n=1}^N \mathcal{L}(f(x_n), y_n) \quad \text{”Empirical risk”}.$$

Trying to make $\mathcal{R}(f)$ small by minimizing $\hat{\mathcal{R}}(f)$ is called **empirical risk minimization** (ERM). ERM is the main idea in this course.

Risk



i Question

Recall our QSAR application.

- ▶ Describe a use case where the IID assumption might be reasonable
- ▶ Describe a use case where the IID assumption might not be reasonable
- ▶ Does minimizing $\mathcal{R}(f)$ guarantee an accurate prediction on any particular x ?

Bibliography